

ПРАКТИКУМ 3 ПРОГРАМУВАННЯ №3
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Завдання №1.

```
#define _USE_MATH_DEFINES
```

```
#include <iostream>
```

```
#include <cmath>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
double taylorSin(double x, double eps) {
```

```
    double term = x, sum = x;
```

```
    int n = 1;
```

```
    while (fabs(term) >= eps) {
```

```
        term *= -x * x / ((2 * n) * (2 * n + 1));
```

```
        sum += term;
```

```
        n++;
```

```
}
```

```
return sum;
```

```
}
```

```
double taylorCos(double x, double eps) {
```

```
    double term = 1.0, sum = 1.0;
```

```
    int n = 1;
```

```
    while (fabs(term) >= eps) {
```

```
        term *= -x * x / ((2 * n - 1) * (2 * n));
```

```
        sum += term;
```

```
        n++;
```

```
    }
```

```
    return sum;
```

```
}
```

```
int main() {

    double x1, x2, dx, eps;

    int choice;


    cout << "x1="; cin >> x1;

    cout << "x2="; cin >> x2;

    cout << "dx="; cin >> dx;

    cout << "Epsilon="; cin >> eps;

    cout << "Function (1=sin, 2=cos): "; cin >> choice;


    cout << fixed << setprecision(6);

    if (choice == 1)

        cout << setw(10) << "----x----" << setw(14) << "--sin_x--"
```

```
<< setw(14) << "--sin_t--" << setw(16) << "-sin_x-sin_t-\n";
```

else

```
cout << setw(10) << "---x---" << setw(14) << "--cos_x--"
```

```
<< setw(14) << "--cos_t--" << setw(16) << "-cos_x-cos_t-\n";
```

```
for (double x = x1; x <= x2 + 1e-9; x += dx) {
```

```
    double rad = x * M_PI / 180.0;
```

```
    double taylor, standard, diff;
```

```
    if (choice == 1) {
```

```
        taylor = taylorSin(rad, eps);
```

```
        standard = sin(rad);
```

```
    }
```

```
    else {
```

```
        taylor = taylorCos(rad, eps);
```

```
standard = cos(rad);
```

```
}
```

```
diff = taylor - standard;
```

```
cout << setw(13) << x
```

```
<< setw(14) << taylor
```

```
<< setw(14) << standard
```

```
<< setw(16) << diff << "\n";
```

```
}
```

```
return 0;
```

```
}
```

```
Консоль отладки Microsoft VI X + -
x1=0
x2=100
dx=10
Epsilon=0.001
Function (1=sin, 2=cos): 2
---x--- --cos_x-- --cos_t-- --cos_x-cos_t--
0.000000 1.000000 1.000000 0.000000
10.000000 0.984808 0.984808 0.000000
20.000000 0.939695 0.939693 0.000003
30.000000 0.866025 0.866025 -0.000000
40.000000 0.766043 0.766044 -0.000001
50.000000 0.642779 0.642783 -0.000003
60.000000 0.500000 0.500000 0.000000
70.000000 0.342022 0.342020 0.000002
80.000000 0.173656 0.173648 0.000008
90.000000 0.000025 0.000000 0.000025
100.000000 -0.173650 -0.173648 -0.000002

C:\Users\kipoc\source\repos\ConsoleApplication14\x64\Debug\ConsoleApplication14.exe (процесс 6312) завершил работу с кодом 0 (0x0).
Нажмите любую клавишу, чтобы закрыть это окно:
```

Завдання №2.

```
#include <iostream>
```

```
#include <cmath>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
// ---- Simple Iteration Method ----
```

```
void solveIteration(double** A, double* b, int n, double eps) {
```

```
    // Check diagonal dominance
```

```
    for (int i = 0; i < n; i++) {
```

```
        double sum = 0;
```

```
        for (int j = 0; j < n; j++)
```

```
            if (j != i) sum += fabs(A[i][j]);
```

```
        if (fabs(A[i][i]) < sum) {
```

```
            cout << "Warning: matrix may not converge (no diagonal dominance on  
row " << i + 1 << ")\\n";
```

```
        }
```

```
    }
```

```
    double* x = new double[n]();
```

```
    double* xNew = new double[n]();
```

```
    int iter = 0;
```

```

do {
    for (int i = 0; i < n; i++) {
        double s = b[i];
        for (int j = 0; j < n; j++)
            if (j != i) s -= A[i][j] * x[j];
        xNew[i] = s / A[i][i];
    }
    double maxDiff = 0;
    for (int i = 0; i < n; i++)
        maxDiff = max(maxDiff, fabs(xNew[i] - x[i]));
    for (int i = 0; i < n; i++)
        x[i] = xNew[i];
    iter++;
    if (maxDiff < eps) break;
    if (iter > 10000) { cout << "Iteration did not converge.\n"; break; }
} while (true);

cout << "\n--- Simple Iteration Method ---\n";
cout << "Iterations: " << iter << "\nResult:\n";
for (int i = 0; i < n; i++)
    cout << "x" << i + 1 << "=" << fixed << setprecision(3) << x[i] << "\n";

delete[] x;
delete[] xNew;
}

// ---- Gauss Elimination Method ----
void solveGauss(double** A, double* b, int n) {
    // Copy arrays
    double** M = new double* [n];
    double* B = new double[n];
    for (int i = 0; i < n; i++) {
        M[i] = new double[n];
        for (int j = 0; j < n; j++) M[i][j] = A[i][j];
        B[i] = b[i];
    }

    // Forward elimination
    for (int i = 0; i < n; i++) {
        // Pivot
        int maxRow = i;
        for (int k = i + 1; k < n; k++)

```

```

        if (fabs(M[k][i]) > fabs(M[maxRow][i])) maxRow = k;
        swap(M[i], M[maxRow]);
        swap(B[i], B[maxRow]);

        for (int k = i + 1; k < n; k++) {
            double factor = M[k][i] / M[i][i];
            for (int j = i; j < n; j++)
                M[k][j] -= factor * M[i][j];
            B[k] -= factor * B[i];
        }
    }

    // Back substitution
    double* x = new double[n];
    for (int i = n - 1; i >= 0; i--) {
        x[i] = B[i];
        for (int j = i + 1; j < n; j++)
            x[i] -= M[i][j] * x[j];
        x[i] /= M[i][i];
    }

    cout << "\n--- Gauss Elimination Method ---\n";
    cout << "Result:\n";
    for (int i = 0; i < n; i++)
        cout << "x" << i + 1 << "=" << fixed << setprecision(3) << x[i] << "\n";

    for (int i = 0; i < n; i++) delete[] M[i];
    delete[] M;
    delete[] B;
    delete[] x;
}

int main() {
    cout << "  A program for solving SLAR\n";
    cout << "SLAR\n";
    cout << "A11X1 + A12X2 + ... +A1nXn = B1\n";
    cout << "A21X1 + A22X2 + ... +A2nXn = B2\n";
    cout << ".....\n";
    cout << "An1X1 + An2X2 + ... +AnnXn = Bn\n\n";

    int n;
    double eps;

```



```
cout << "Enter the size:"; cin >> n;  
cout << "Epsilon:"; cin >> eps;
```

```
double** A = new double* [n];  
for (int i = 0; i < n; i++) A[i] = new double[n];  
double* b = new double[n];
```

```
cout << "\nEnter coefficients:\n";  
for (int i = 0; i < n; i++)  
    for (int j = 0; j < n; j++) {  
        cout << "A" << i + 1 << j + 1 << ":"; cin >> A[i][j];  
    }
```

```
cout << "\nEnter free members:\n";  
for (int i = 0; i < n; i++) {  
    cout << "B" << i + 1 << ":"; cin >> b[i];  
}
```

```
solveIteration(A, b, n, eps);  
solveGauss(A, b, n);
```

```
for (int i = 0; i < n; i++) delete[] A[i];  
delete[] A;  
delete[] b;  
return 0;  
}
```

```
Консоль отладки Microsoft Vi X + v
A program for solving SLAR
SLAR
A11X1 + A12X2 + ... +A1nXn = B1
A21X1 + A22X2 + ... +A2nXn = B2
.....
An1X1 + An2X2 + ... +AnnXn = Bn

Enter the size:4
Epsilon:42

Enter coefficients:
A11:34
A12:3
A13:2
A14:3
A21:4
A22:5
A23:12
A24:3
A31:5
A32:34
A33:23
A34:21
A41:2
A42:3
A43:4
A44:3

Enter free members:
B1:4
B2:23
B3:23
B4:4
Warning: matrix may not converge (no diagonal dominance on row 2)
Warning: matrix may not converge (no diagonal dominance on row 3)
Warning: matrix may not converge (no diagonal dominance on row 4)

--- Simple Iteration Method ---
Iterations: 1
Result:
x1=0.118
x2=4.600
x3=1.000
x4=1.333

--- Gauss Elimination Method ---
Result:
x1=0.138
x2=0.557
x3=2.201
x4=-2.251

C:\Users\kipoc\source\repos\ConsoleApplication14\x64\Debug\ConsoleApplication14.exe (процесс 18120) завершил работу с кодом 0 (0x0).
Нажмите любую клавишу, чтобы закрыть это окно:
```